$0, 1, \ldots, \nu$. (This is a simplifying assumption the removal of which would require somewhat more intricate assumptions regarding the differential equations.)

With these assumptions in hand, begin from (2.9) and write

$$z_{n+1} - z(t_{n+1}) = \mathscr{G}_h(z_n) - \mathscr{G}_h(z(t_n)) + \mathscr{G}_h(z(t_n)) - \mathscr{F}_h(z(t_n)),$$

then take norms and use the triangle inequality and (2.10), (2.11) to get the following recurrent inequality for the error $\varepsilon_n = \|z_n - z(t_n)\|$:

$$\varepsilon_{n+1} \leq (1 + Lh)\varepsilon_n + \bar{K}h^{p+1}.$$

From this, the bound

$$\varepsilon_n \leq \frac{\bar{K}}{L} e^{Lnh} h^p, \qquad n = 0, 1, \ldots \nu, \tag{2.12}$$

follows by a straightforward calculation, for $h \leq \Delta$.

The assumption (2.10) that the local error is of order $p + 1$, $p > 0$, is termed the *consistency* of the numerical method. We say the method is *consistent of order p*.

The assumption (2.11) that the method does not increase the separation between two nearby trajectories by more than a factor of the form $1 + hL$ in each step is referred to as the *stability* of the method.

This result shows that a method which is consistent of order p and stable is convergent of order p.

Example 2.2 (The Verlet Method is 2nd Order) In this example, assume a single degree of freedom system, i.e. $q, p \in \mathbb{R}$, and take $M = 1$. The Verlet method can be written in the form of a map, as in (2.8), or, in slightly more detail, as

$$Q = q + hp + \frac{h^2}{2} F(q), \tag{2.13}$$

$$P = p + \frac{h}{2}\left[F(q) + F(q + hp + \frac{h^2}{2}F(q))\right]. \tag{2.14}$$

The first equation is already a polynomial, i.e. it is in the form of a series expansion in powers of h where the coefficients are functions of the starting point (q, p). The second equation may be written as a series expansion in powers of h as well:

$$P = p + \frac{h}{2}F(q) + \frac{h}{2}\left[F(q) + hF'(q)(p + \frac{h}{2}F(q)) + \frac{h^2}{2}F''(q)(p + \frac{h}{2}F(q))^2 + \ldots\right].$$

Note that the neglected terms will involve 4th (and higher) powers of h.

Combining terms of like powers of h, we have

$$P = p + hF + \frac{h^2}{2}pF' + \frac{h^3}{4}\left[F'F + p^2F''\right] + \mathcal{O}(h^4).$$

We carry these terms to compare against the expansion of the exact solution. Since $\dot{q} = p$, we have $\ddot{q} = \dot{p} = F(q)$, and the third derivative is $q^{(3)} = F'(q)p$. On the other hand $\dot{p} = F(q)$ implies that $\ddot{p} = F'(q)\dot{q} = pF'$, and thus

$$p^{(3)} = p^2F'' + F'F.$$

The Taylor expansion of the solution is (taking $q(t) = q, p(t) = p$):

$$q(t+h) = q + hp + \frac{h^2}{2}F + \frac{h^3}{6}F'p + \mathcal{O}(h^4),$$
$$p(t+h) = p + hF + \frac{h^2}{2}pF' + \frac{h^3}{6}\left[p^2F'' + F'F\right] + \mathcal{O}(h^4).$$

We now examine the series expansions for the exact and Verlet solutions and find that these differ in the third (and higher) order terms.

$$Q - q(t+h) = \frac{h^3}{6}F'p + \mathcal{O}(h^4),$$

and

$$P - p(t+h) = \frac{h^3}{12}\left[p^2F'' + F'F\right] + \mathcal{O}(h^4).$$

These relations can be summarized as telling us that

$$\|\mathscr{G}_h(\boldsymbol{z}) - \mathscr{F}_h(\boldsymbol{z})\| = \kappa(\boldsymbol{z})h^3 + \mathcal{O}(h^4),$$

where $\kappa(\boldsymbol{z}) = \kappa(q,p)$ is a function of the position and momentum.

We may then define

$$\bar{K} = \max_{t\in[0,\tau]} \kappa(\boldsymbol{z}(t))$$

bounding the local error by (with neglect of the fourth order terms) $\bar{K}h^3$. Thus the Verlet method is consistent of order two.

To complete the convergence proof for Verlet's method, we would still need to verify the second assumption. This requires the assumption that the force field $\boldsymbol{F}$ satisfy a Lipschitz condition:

$$\|\boldsymbol{F}(\boldsymbol{u}) - \boldsymbol{F}(\boldsymbol{w})\| \leq \hat{L}\|\boldsymbol{u} - \boldsymbol{w}\| \tag{2.15}$$